

# Sai Praneet Reddy Chinthala

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## EDUCATION

### George Mason University

M.S. Data Analytics Engineering – GPA: 3.97/4.0

Fairfax, Virginia

Aug 2024 – May 2026

### Mahindra University

B.Tech, Electronics and Computer Engineering

Hyderabad, India

Sept 2020 – May 2024

## TECHNICAL SKILLS

**Languages:** Python, SQL, R

**GenAI & ML:** RAG, LLMs, prompt engineering, embeddings, vector search, LangChain, LangGraph, FAISS, ChromaDB, BM25, Cohere Rerank, Langfuse, NLP, SBERT, PyTorch, scikit-learn, XGBoost

**Data Engineering:** ETL/ELT pipelines, data validation & quality, Apache Spark, PySpark, Databricks, Hadoop, pandas, NumPy, MySQL, MongoDB, Power BI

**Cloud & APIs:** AWS (S3, EC2, Lambda, DynamoDB), Azure OpenAI, Docker, CI/CD (GitHub Actions), Git, FastAPI, REST APIs

## EXPERIENCE

### Data Engineer (Volunteer)

Aug 2026 – Present

Saayam For All – 501(c)(3) non-profit

Remote

- Support the data-analytics workstream with ETL and data-quality tooling for BI reporting.
- Built a Python synthetic-data generator and validator for **10 relationally consistent tables** (schemas matched to production; foreign-key, geo-coordinate and fake-PII checks), enabling dashboard work **without real user data**.
- Built an AWS Lambda analytics API (Python, pandas) serving **rating-distribution and organization-growth trend charts** for the organization dashboard, with 7-day to custom time windows, country filtering and input validation.

### Graduate Teaching Assistant (AIT 614 – Big Data Essentials)

Jan 2026 – May 2026

George Mason University

Fairfax, Virginia

- Led **10 hands-on labs** across **two sections** on Databricks, Spark/PySpark, MongoDB, Hadoop, AWS and LangChain RAG.
- Graded **100+ Python and PySpark submissions** from 35 students on ETL, model training and feature engineering.
- Solved the professor-assigned problem sets ahead of the course calendar, producing **the reference solutions used by other TAs**, and guided students through questions and term-long big-data projects.

### Generative AI Engineer Intern

Feb 2024 – May 2024

Mahindra & Mahindra Ltd

Remote

- Consolidated **20+ machine-breakdown root-cause Excel logs** from three plants into one standardized dataset with pandas.
- Adapted an internal Azure OpenAI RAG chatbot (embeddings + vector similarity search) to the cleaned corpus so **engineers could query breakdown history in natural language**.

## RESEARCH

### Fair Bilevel Optimization for Collaborative Classification

Jan 2026 – Present

George Mason University, Dept. of Information Sciences & Technology

Fairfax, Virginia

- Developing a federated bilevel optimization method (Augmented Lagrangian) that trains fairness-constrained classifiers across institutions without sharing raw data.

## PROJECTS

### DAPSE: Arctic Policy Intelligence Engine

Jan 2026 – May 2026

Master's Capstone (team of 6) – Product Owner

Code private

- Co-developed a RAG engine with BM25 + FAISS hybrid retrieval, Reciprocal Rank Fusion and Cohere Rerank over **955+ documents; nDCG@5 0.832, Precision@5 0.954**.
- Directed delivery of a verify-then-escalate agent loop and an **8-dimension risk-scoring engine**, with Langfuse observability and Docker deployment.

### Senior Care RAG Agent | [GitHub](#)

Aug 2026

- Built a LangGraph agentic RAG assistant (query → retrieve → grade → generate → verify); every answer grounded and cited.
- 54-question eval set: **Hit@4 0.926, MRR 0.861, faithfulness 0.778** (independently LLM-judged); served via FastAPI, Streamlit and an MCP server, with Docker and CI.

### Bridges-NBI-Analysis | [GitHub](#)

Aug 2025 – Dec 2025

- Built a PySpark ETL + ML pipeline over **950K+ National Bridge Inventory records**; engineered **26 features from 15 source items** and handled a 9:1 class imbalance.
- Tuned **XGBoost from 0.75 to 0.93 ROC-AUC** and identified **truck traffic as the strongest predictor**; persisted to MongoDB with automated Databricks SQL reporting.